Data audit

RESPONSIBLE AI DATA MANAGEMENT

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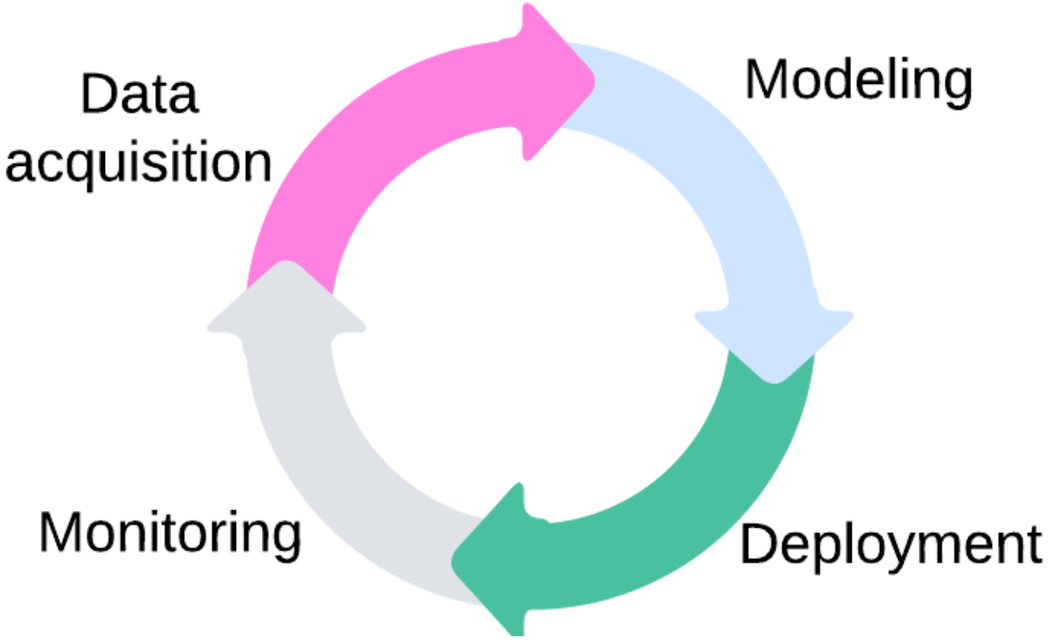
datacamp

What we will cover

- Data auditing
- Data validation
- Bias mitigation strategies

Data audit

- Complete data review throughout a project's lifecycle
- Check its technical and responsible state
- Conducted after a major change to the project
- Updated documentation for audit



Performing data audits

- Done frequently
- Ensure issues are not unintentionally introduced
- Safeguard from complications and amplifying errors
- Remain transparent and accountable
- Build trust

AI Financial advisor

- Behavior profile and plan goals
- Develop a strategy
- Match goals with investment products
- Create investment portfolio

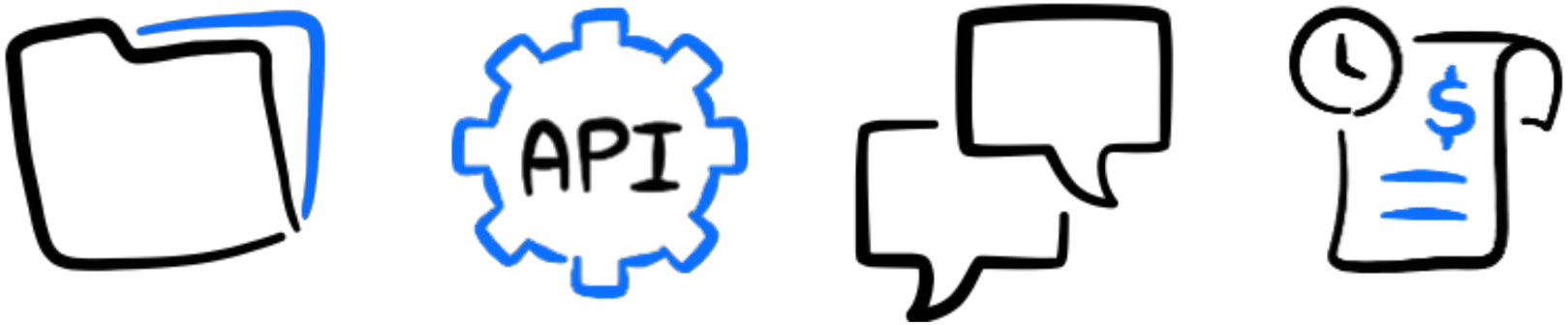
Data sources:

- User data from chat and uploads
- External real-time and historical data (Bloomberg)

[1] Image by Streamline HQ

Project data sources

- User provided data
- External data from API

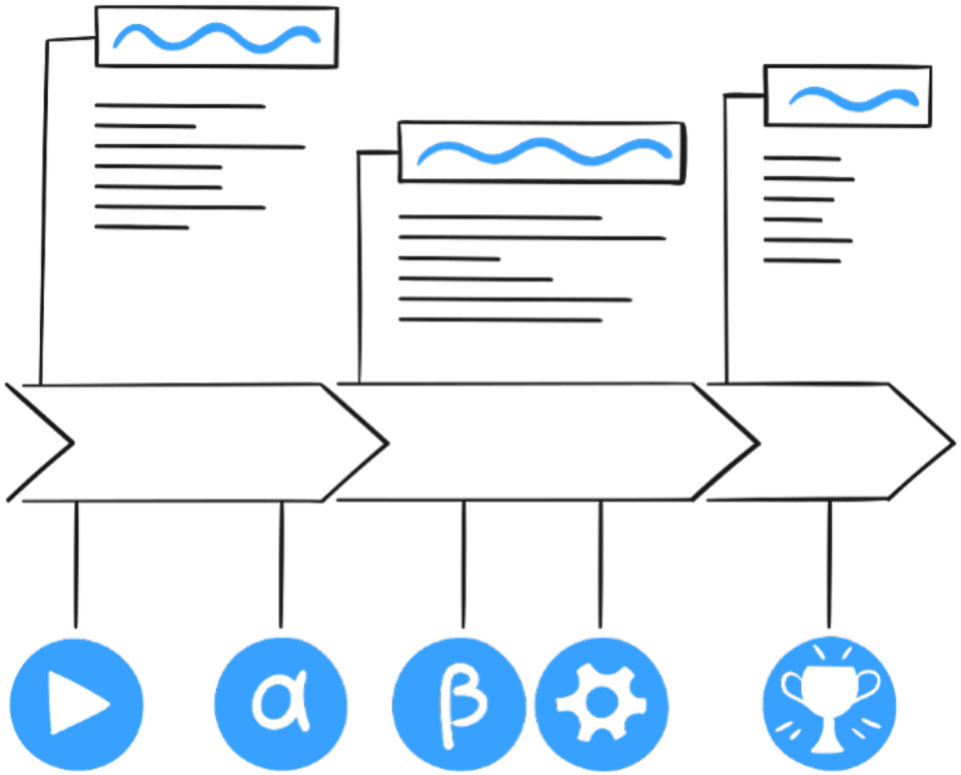


[1] Image by Streamline HQ

Data audit setup

Use Data Management Plan:

- Details of data
- Audit frequency
- Required tests
- Assigned team members



[1] Image by Streamline HQ

Data audits schedule

- Initial data exploration
- Regular audits for preprocessing and modeling
- Any substantial data changes

[1] Image by Streamline HQ

Ongoing data audits

- Regular continuous monitoring
- Data quality and compliance
- Model performance and fairness metrics
- Data usage and storage
- Security and scalability

Model drift

- Model may become less accurate with time
- Societal changes, market conditions, or underlying variables
- Detect model drift by logging predictions and assessing performance metrics
- Alert if metrics breach threshold
- Fix by checking for new data or retraining the model

AI financial advisor data audits

- Initial data exploration
- Preprocessing with transformations and cleaning
- Modeling with fairness assessment
- Pre-deployment audit
- Continuous monitoring

[1] Image by Streamline HQ

Let's practice!

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Data validation

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What we will cover

- Data validation
- Responsible dimensions
- AI financial advisor

Data validation

- More granular approach
- Check data's technical integrity
- Check fairness

Technical integrity

- Complete data
- No duplicates, errors, or outdated data
- Legally compliant
- Checks for accuracy, consistency, completeness, and timeliness

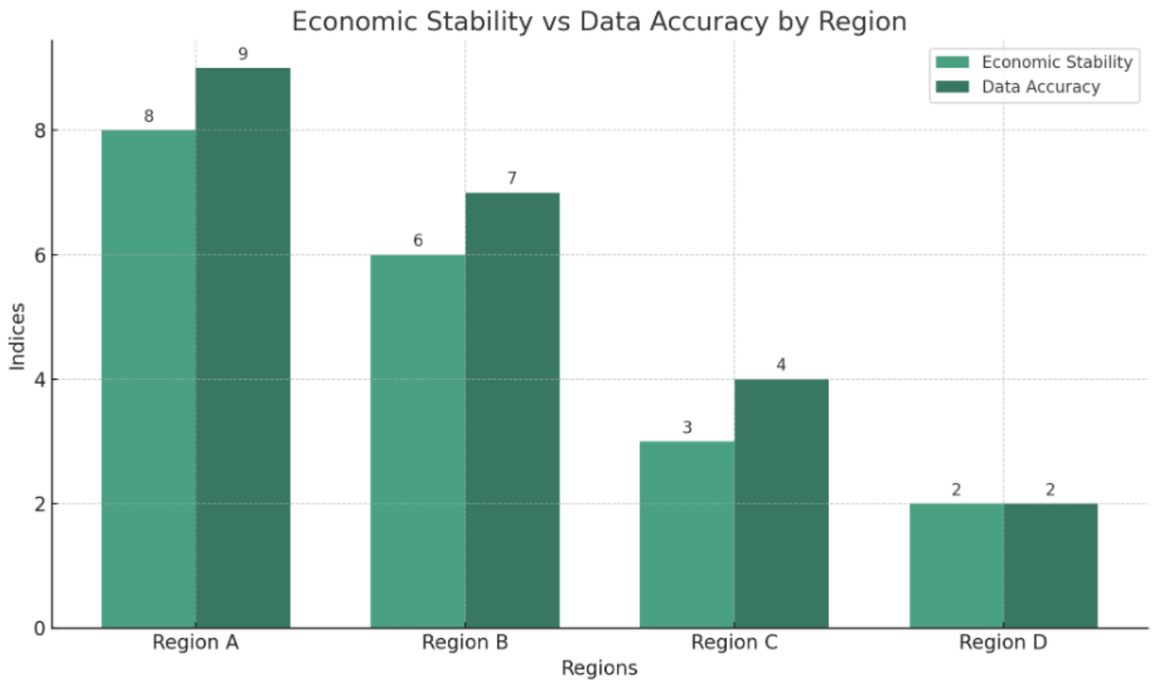
Financial advisor: technical integrity

- Discrepancies and anomalies in data
- Incorrectly assigned fields
- Spot and correct early

User Name	Income	Income Category
John Doe	3500	Salary
Jane Smith	4200	Salary
Alex Johnson	3900	Bonus
Maria Garcia	4500	Expense

Financial advisor: technical integrity

- Anomalies in data collection periods and methods
- Regional economic disparities
- Not optimized for all user groups



Fairness assessment

- Equal opportunity
- Disparate impact
- Demographic parity
- Applied to all project stages

Financial advisor:

- All individuals receive relevant and beneficial advice
- Uniform recommendations across all demographic groups
- No group is disproportionately affected

Data validation approaches

- Identify key variables
- Analyze data distribution
- Clean the data and apply statistical tests
- Balance the data
- Check fairness metrics
- Test models with diverse data sets

Financial advisor

- Remove outliers and impute missing values
- Validate with descriptive statistics before and after
- Oversample for low-income group
- Check model performance and fairness metrics
- Cross-validation with stratification
- Test on "unseen" data

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Data validation best practices

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What we will cover

- Subgroup analysis
- Missing values
- Outlier removal
- Correcting data inconsistencies
- Feature scaling
- Feature encoding
- Dimensionality reduction

Subgroup analysis

Divide into subgroups based on protected characteristics

Evaluate each subgroup's statistical distribution and model performance

Evaluate model fairness metrics for each subgroup

Apply mitigation strategies

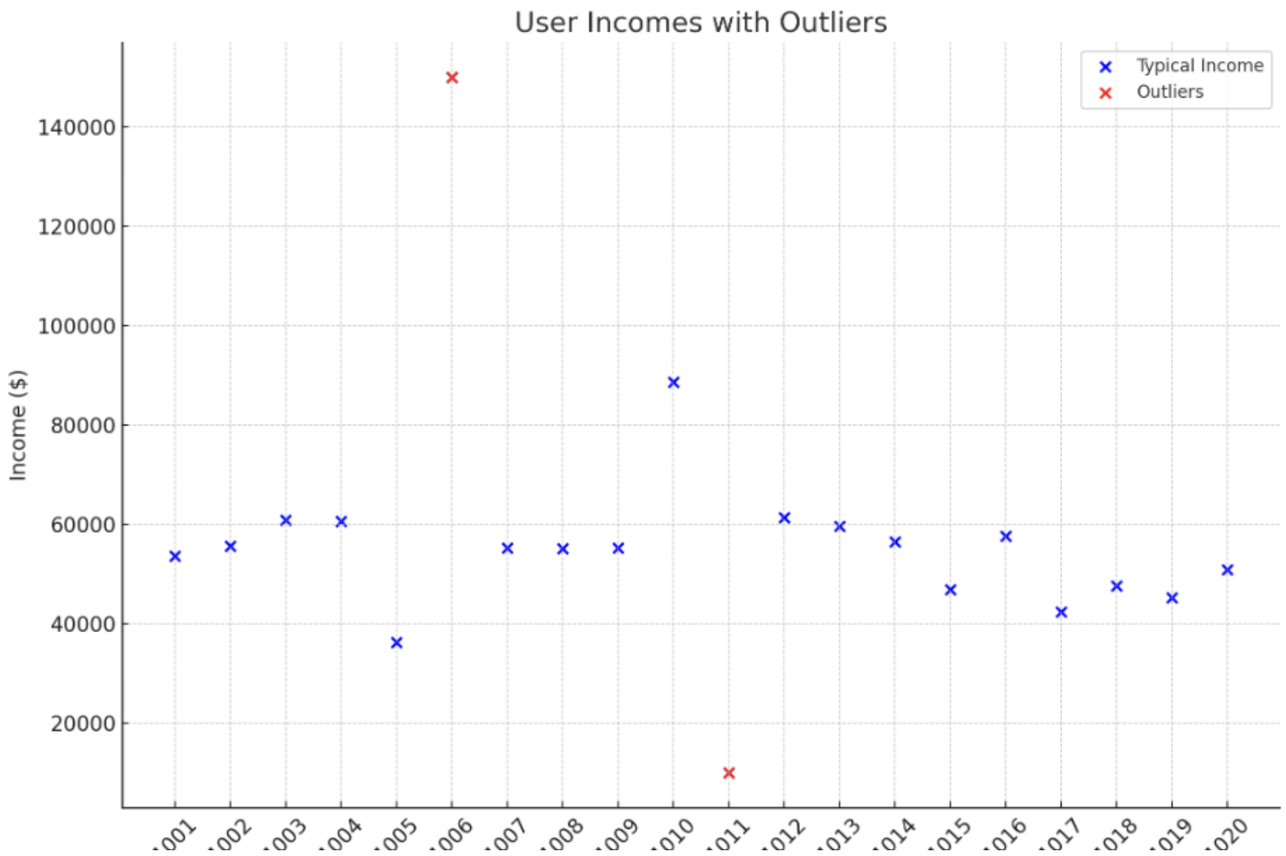
Missing data

- Common in large datasets
- Data deletion
- Imputation strategies and model-based approaches
- Subgroup analysis for validation

User ID	Name	Email	Phone Number	Investment Preference
101	John Doe	johndoe@example.com		Stocks
102	Jane Smith		555-0199	Bonds
103		carolb@example.com	555-0187	
104	Mike Johns	mikej@example.com		Real Estate
105	Emily White		555-0133	Cryptocurrency

Outlier removal

- Statistical methods like z-scores and IQR, or robust scaling
- Validate for fair treatment across data segments

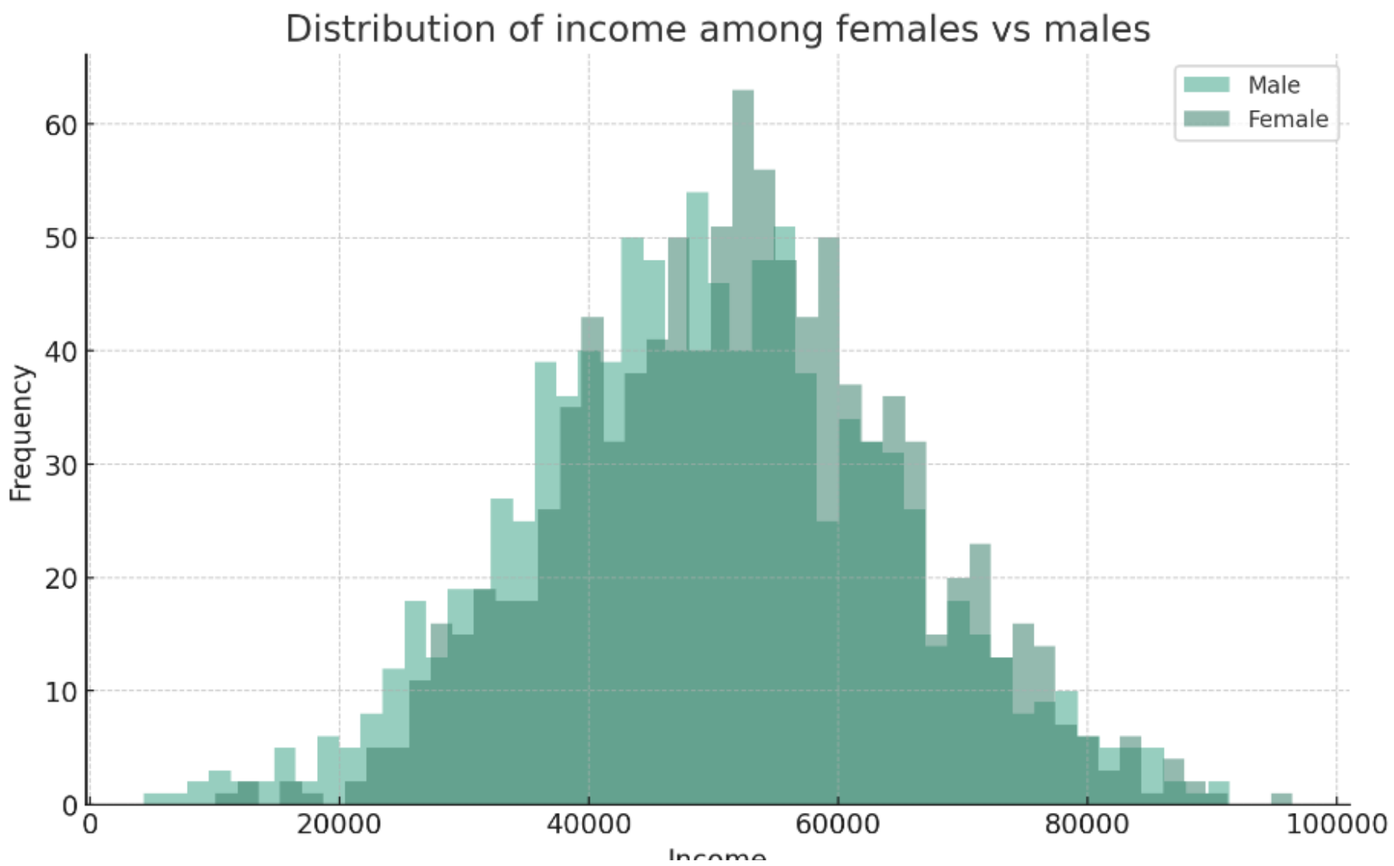


Data inconsistencies

- Data quality affects model integrity and reliability
- Data standardization and validation rules
- Subgroup normalization

Feature scaling

- Feature scaling to transform input features
- Validate by checking the distributions among groups

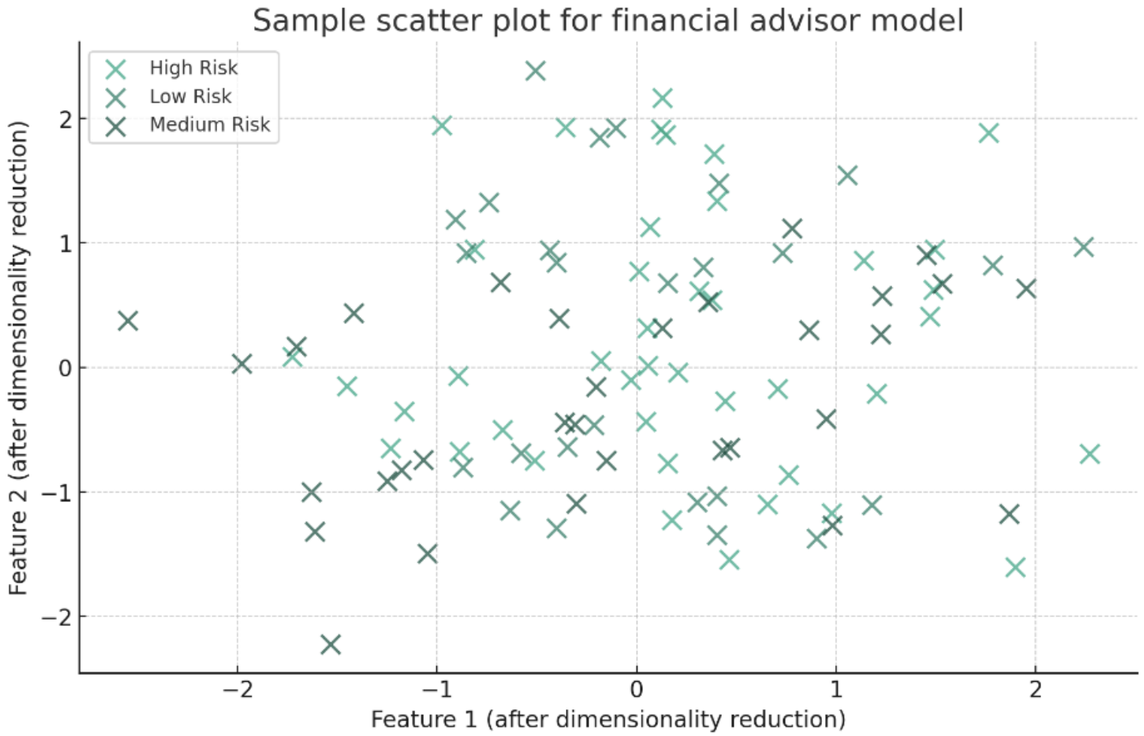


Feature encoding

- Assess encoding effect on outcomes
- Check for biases and information loss
- Check for overfitting
- Employ regularization and dimensionality reduction

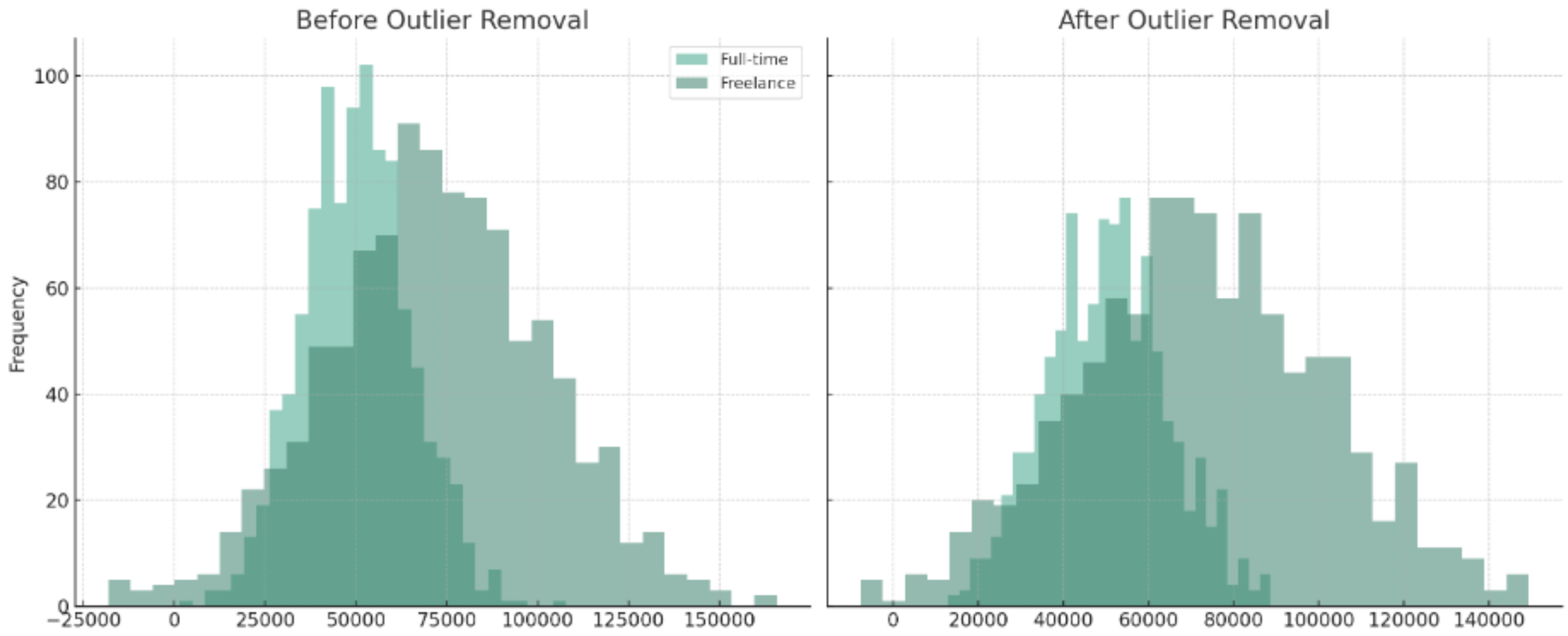
Dimensionality reduction

- Reduce input features and preserve essential information
- May create bias
- Use fairness-conscious techniques like t-SNE



Financial advisor

- "Annual income" and "Investment frequency" features
- Adjust outliers and scale
- Subgroup analysis



Let's practice!

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Bias mitigation

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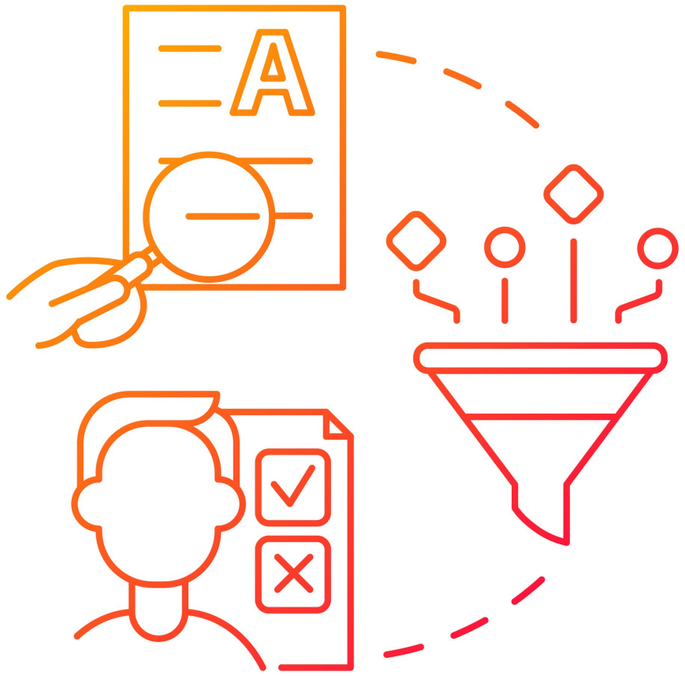
We will cover

- Strategies throughout the project lifecycle
- Re-sampling and re-weighting
- Re-labeling and sensitivity attribute removal
- Adversarial training and model re-calibration
- User behavior
- Fairness constraints and algorithm selection
- Other strategies exist

Bias revisited

- Systematic errors or prejudices
- Unfair outcomes for groups or individuals

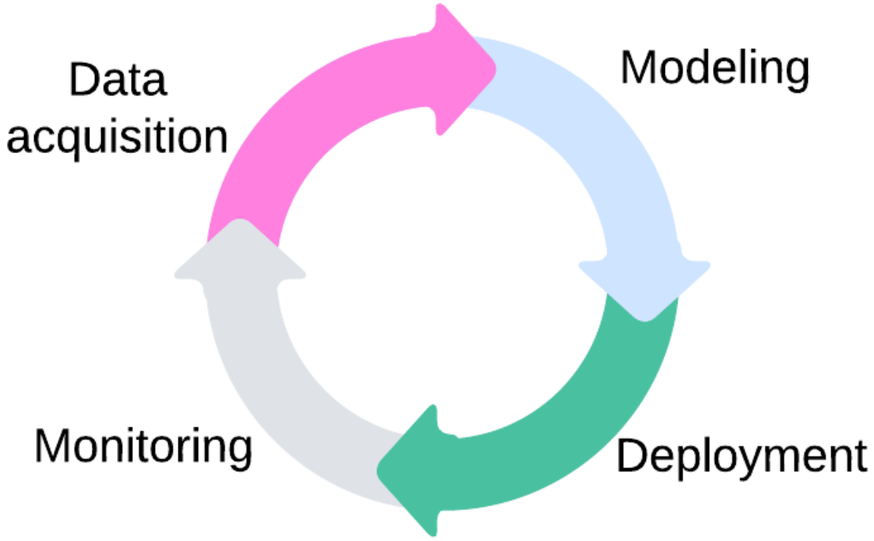


BIAS

[1] Image by Streamline HQ

Bias across project

- In raw data
- In pre-processing
- In modeling
- In post-deployment



Bias in pre-processing

- Re-sampling
- Re-weighting
- Re-labeling
- Sensitivity attribute removal

Strategies in pre-processing

- Re-sampling:
  - Adjust distribution for balanced class representation
- Re-weighting:
  - Correct biases with class weights, keeping dataset size constant
- Re-labelling:
  - Update labels for accurate target representation
- Sensitivity attribute removal:
  - Eliminate protected attributes

Bias in modeling

- Bias from the algorithm
- Adversarial training
- Fairness constraints
- Algorithm selection with simpler, robust, or generalizable models

Financial advisor

- Bias related to regional economic disparities
- Adversarial training to correct bias
- Decision trees and regularized linear models

[1] Image by Streamline HQ

Bias in post-deployment and use

- Monitor bias to prevent model drift
- Model re-calibration
- Improper application use

[1] Image by Streamline HQ

Financial advisor

Model drift:

- Underrepresented freelance economy

Incorrect use:

- Updating financial information

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Congratulations!

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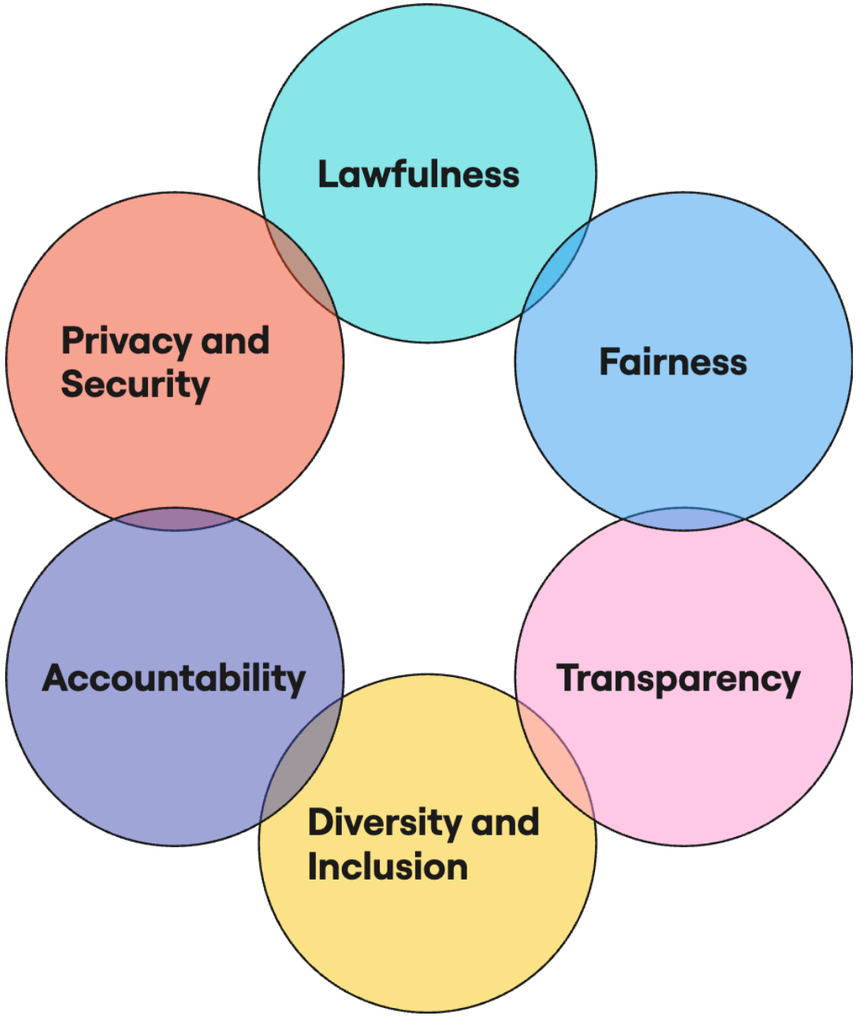
Chapter 1: Responsible data dimensions

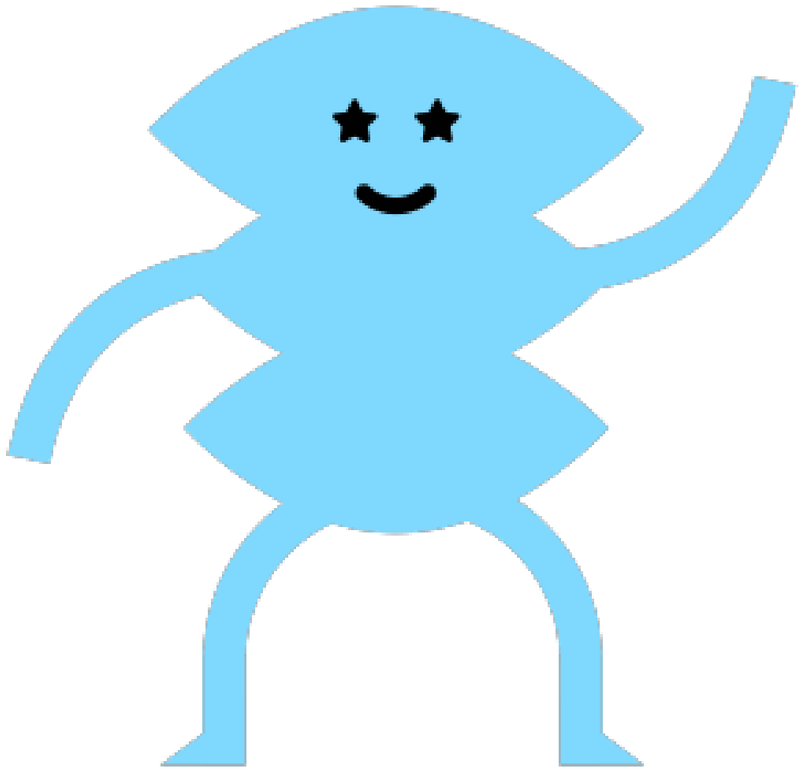


Chapter 2: Lawfulness and compliance

- Country/industry specific laws
- Data owner's rights
- Informed consent
- Data use agreements
- Data licensing

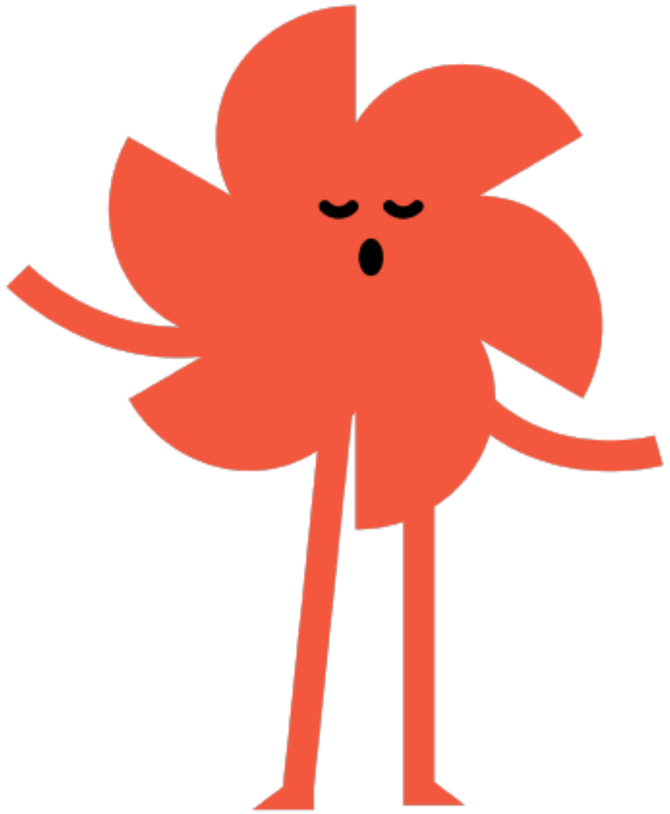
Chapter 3: Data sources and integration

- Data sources
- Diversity and representation
- Bias and fairness



Chapter 4: Validation and bias

- Bias and fairness
- Data preprocessing, modeling and post-deployment
- Fairness approaches



What's next?

Start your responsible AI journey!

Congratulations!

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